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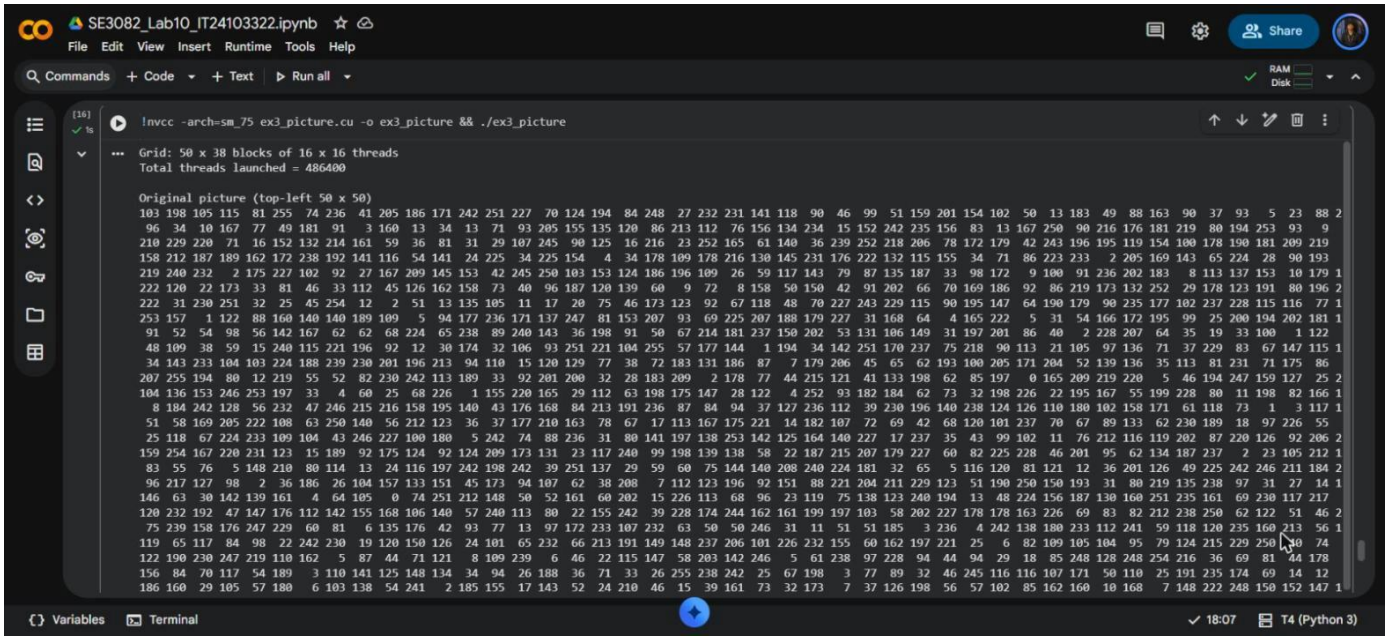
BSc (Hons) in Computer Science
Year 3

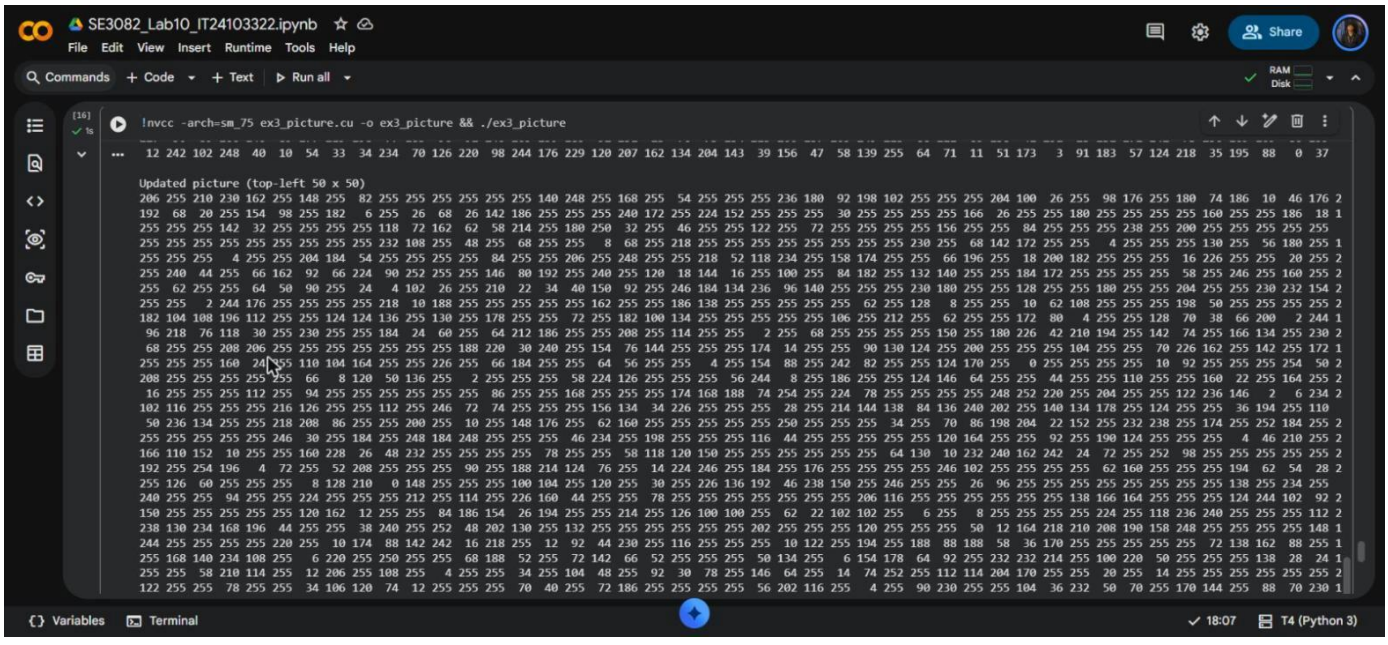
Lab Sheets

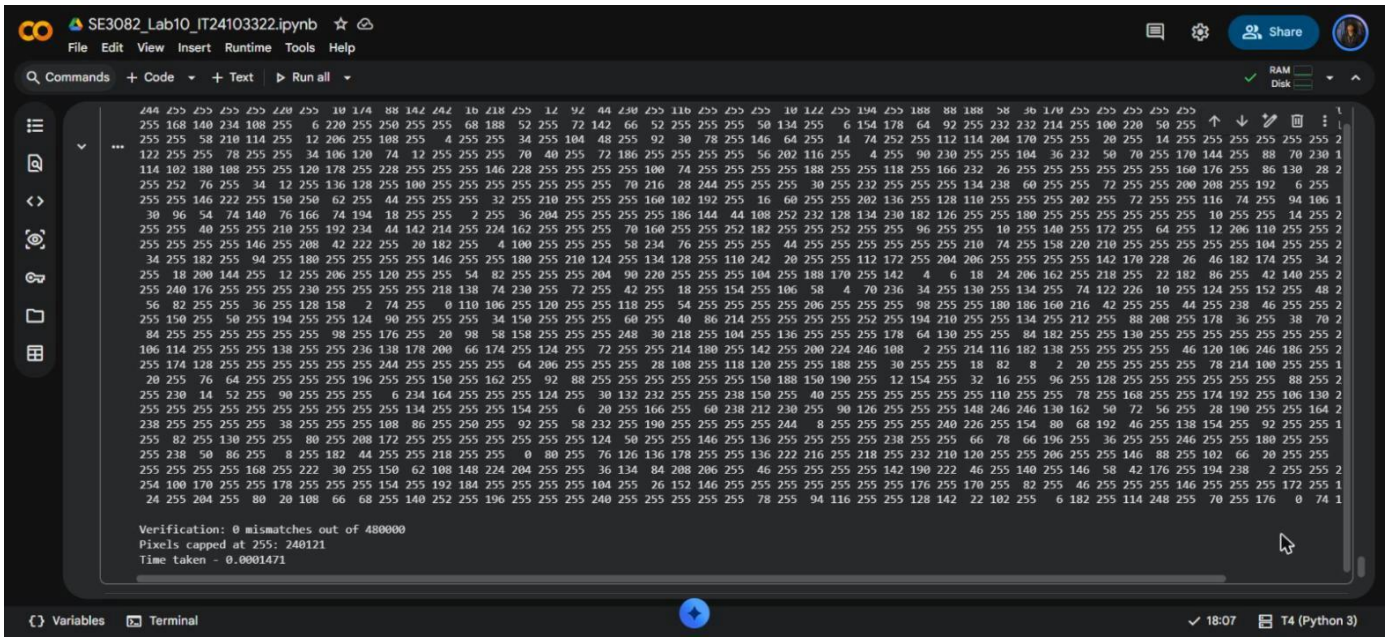
SE3082 – Parallel Computing **Semster 1, 2026**

IT24103322

Exercise 3







The 800 × 600 picture is one flat array of 480,000 unsigned char values, filled with rand() % 256. dim3 threadsPerBlock(16, 16) gives 256 threads per block, and the grid is rounded up in both directions: 800 / 16 = 50 blocks across and 600 / 16 = 37.5 → 38 blocks down. The PictureKernel uses Row from y and Col from x, works on pixel Row * width + Col only if both are inside the picture, and doubles the value with a cap at 255. The top-left 50 × 50 pixels were printed before and after.

Item	Expected value	My result
Grid	50 x 38 blocks of 16 x 16 threads	50 x 38 blocks of 16 x 16 threads
Total threads launched	486,400	486400
First ten original values	103 198 105 115 81 255 74 236 41 205	103 198 105 115 81 255 74 236 41 205
First ten updated values	206 255 210 230 162 255 148 255 82 255	206 255 210 230 162 255 148 255 82 255
Verification	0 mismatches out of 480000	0 mismatches out of 480000
Pixels capped at 255	about half (240,121)	240121

1. **Roughly how many of the 480,000 pixels will be capped at 255?**

   About half. Every original value from 128 to 255 becomes 256 or more when doubled, and that is half of the 256 possible values. 240,121 pixels (50.0 %) were capped. Without the cap 2 × 200 = 400 would wrap around to 144 in an unsigned char.